indiv_region_increases

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This script test whether non-EUCA per-capita arrest rates rose more than EUCA per-capita rates in all states and AORs and with all denominators

```
aggregate_data <- function(data,
                           cols_to_keep,
                           pop_col,
                           agg_noneuca = TRUE){


  pop_cols <- c("pop_ACS", "pop_MPI")

  data_pop <- data %>%
    select(-setdiff(pop_cols, pop_col)) %>%
    rename("pop" = !!sym(pop_col))

  raw_pop <- data_pop %>%
    group_by(aor, region) %>%
    summarize(pop = first(pop), .groups = "drop")

  # aggregate regions if reuqested
  if(agg_noneuca){
    data_pop <- data_pop %>%
      mutate(region = ifelse(region == "EUCA", "EUCA", "nonEUCA"))

    raw_pop <- raw_pop %>%
      mutate(region = ifelse(region == "EUCA", "EUCA", "nonEUCA")) %>%
      group_by(aor, region) %>%
      summarize(pop = sum(pop), .groups = "drop")
  }

  # checksum value
  checksum <- data_pop %>%
    group_by(aor, region, pop) %>%
    slice_head(n=1) %>%
    ungroup() %>%
    group_by(region) %>%
    summarize(pop_checksum = sum(pop), .groups = "drop")

  # find geographic keys
  geo_keys <- intersect(cols_to_keep, c("aor", "region"))

  # population map
  pop_map <- data_pop %>%
```

```r
    group_by(aor, region, pop) %>%
    slice_head(n = 1) %>%
    ungroup() %>%
    group_by_at(geo_keys) %>%
    summarize(pop = sum(pop), .groups = "drop")

  # aggregate arrests, join population back in
  data_pop <- data_pop %>%
    group_by_at(cols_to_keep) %>%
    summarize(n_arrests = sum(n_arrests, na.rm = TRUE), .groups = "drop") %>%
    left_join(pop_map, by = geo_keys) %>%
    mutate(ratio = n_arrests/pop*1000,
           log_pop = log(pop))

  # checksum
  total_aggregated <- data_pop %>%
    group_by_at(geo_keys) %>%
    summarize(pop = first(pop), .groups = "drop") %>%
    pull(pop) %>%
    sum(na.rm = TRUE)
  total_checksum <- sum(checksum$pop_checksum, na.rm = TRUE)
  if (!(dplyr::near(total_checksum, total_aggregated))) {
    warning("Checksum Failed!")
  }

  return(data_pop)
}
```

MPI, AOR

```r
trajectory <- read_csv("../data/glm_trajectory_type_threatlevel_convicted_ACS_MPI_AOR.csv")
```

```
## Rows: 113400 Columns: 10
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## chr (4): aor, region, method, alt_method
## dbl (6): window, convicted, threat_level, n_arrests, pop_ACS, pop_MPI
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
trajectory_agg <- aggregate_data(trajectory, c("region", "aor", "window"), "pop_MPI", TRUE) %>%
  mutate(window = ifelse(window < 0, -1, 0)) %>%
  group_by(window, region, aor, pop) %>%
  summarize(n_arrests = sum(n_arrests)) %>%
  mutate(ratio = n_arrests/pop *1000)
```

```
## `summarise()` has regrouped the output.
## i Summaries were computed grouped by window, region, aor, and pop.
## i Output is grouped by window, region, and aor.
## i Use `summarise(.groups = "drop_last")` to silence this message.
## i Use `summarise(.by = c(window, region, aor, pop))` for per-operation grouping
##   (`?dplyr::dplyr_by`) instead.
```

```r
trajectory_agg_euca <- trajectory_agg %>%
  filter(region == "EUCA") %>%
```

```
  pivot_wider(id_cols = aor, names_from = window, values_from = ratio) %>%
  mutate(diff_euca = `0`-`-1`) %>%
  mutate(region = "EUCA") %>%
  select(diff_euca, aor)

trajectory_agg %>%
  filter(region == "nonEUCA") %>%
  pivot_wider(id_cols = aor, names_from = window, values_from = ratio) %>%
  mutate(diff_noneuca = `0`-`-1`) %>%
  mutate(region = "Non-EUCA") %>%
  select(diff_noneuca, aor) %>%
  merge(trajectory_agg_euca) %>%
  mutate(diff = diff_noneuca - diff_euca) %>%
  arrange(diff)
```

```
##               aor diff_noneuca diff_euca      diff
## 1   San Francisco     4.264320  2.269430  1.994890
## 2     Los Angeles     8.276741  5.120844  3.155898
## 3         Seattle     9.851351  1.485294  8.366057
## 4   New York City    11.419123  2.401267  9.017856
## 5          Boston    12.118928  2.120000  9.998928
## 6       Baltimore    13.466480  3.066667 10.399814
## 7          Newark    16.863535  5.655172 11.208362
## 8         Buffalo    14.383401  2.800355 11.583046
## 9          Dallas    21.485230  9.854304 11.630926
## 10        Chicago    15.004410  3.041237 11.963173
## 11        Atlanta    19.037661  2.491803 16.545858
## 12      San Diego    39.628597 23.037897 16.590700
## 13 Salt Lake City    21.313725  4.000000 17.313725
## 14         Denver    21.175115  3.789474 17.385642
## 15        Houston    25.688052  7.313081 18.374971
## 16          Miami    24.487424  5.859155 18.628269
## 17        Phoenix    26.355172  2.533333 23.821839
## 18     Washington    30.188011  3.791667 26.396344
## 19   Philadelphia    37.107143  5.513514 31.593629
## 20    San Antonio    39.622596  6.199438 33.423158
## 21      St. Paul     42.015075  4.263158 37.751917
## 22    New Orleans    46.310185  4.366667 41.943519
## 23        Detroit    49.284722  2.807692 46.477030
## 24        El Paso    70.740809 10.112767 60.628041
## 25      Harlingen    95.774231  4.821006 90.953224
```

ACS, AOR

```
trajectory <- read_csv("../data/glm_trajectory_type_threatlevel_convicted_ACS_MPI_AOR.csv")
```

```
## Rows: 113400 Columns: 10
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## chr (4): aor, region, method, alt_method
## dbl (6): window, convicted, threat_level, n_arrests, pop_ACS, pop_MPI
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
trajectory_agg <- aggregate_data(trajectory, c("region", "aor", "window"), "pop_ACS", TRUE) %>%
  mutate(window = ifelse(window < 0, -1, 0)) %>%
  group_by(window, region, aor, pop) %>%
  summarize(n_arrests = sum(n_arrests)) %>%
  mutate(ratio = n_arrests/pop *1000)
```

```
## `summarise()` has regrouped the output.
## i Summaries were computed grouped by window, region, aor, and pop.
## i Output is grouped by window, region, and aor.
## i Use `summarise(.groups = "drop_last")` to silence this message.
## i Use `summarise(.by = c(window, region, aor, pop))` for per-operation grouping
##   (`?dplyr::dplyr_by`) instead.
```

```r
trajectory_agg_euca <- trajectory_agg %>%
  filter(region == "EUCA") %>%
  pivot_wider(id_cols = aor, names_from = window, values_from = ratio) %>%
  mutate(diff_euca = `0`-`-1`) %>%
  mutate(region = "EUCA") %>%
  select(diff_euca, aor)

trajectory_agg %>%
  filter(region == "nonEUCA") %>%
  pivot_wider(id_cols = aor, names_from = window, values_from = ratio) %>%
  mutate(diff_noneuca = `0`-`-1`) %>%
  mutate(region = "Non-EUCA") %>%
  select(diff_noneuca, aor) %>%
  merge(trajectory_agg_euca) %>%
  mutate(diff = diff_noneuca - diff_euca) %>%
  arrange(diff)
```

```
##                aor diff_noneuca diff_euca      diff
## 1    San Francisco     1.315308 0.2973953  1.017913
## 2      Los Angeles     2.191039 0.7735320  1.417507
## 3    New York City     1.706730 0.2240765  1.482653
## 4          Seattle     3.356949 0.3009212  3.056027
## 5           Newark     4.035230 0.5227791  3.512451
## 6           Boston     4.425621 0.3265075  4.099113
## 7          Chicago     5.173567 0.5013434  4.672223
## 8        Baltimore     5.299264 0.4840831  4.815181
## 9            Miami     6.822478 0.7483105  6.074168
## 10         Detroit     6.947925 0.2615887  6.686337
## 11          Dallas     9.519257 1.7399275  7.779330
## 12         Houston     8.940948 0.9434457  7.997502
## 13 Salt Lake City     8.712429 0.7096074  8.002822
## 14         Atlanta     9.164228 0.5172707  8.646957
## 15          Denver     9.660708 0.7106409  8.950067
## 16         Phoenix     9.523976 0.2855318  9.238444
## 17    Philadelphia    10.529848 0.9788445  9.551004
## 18      Washington    10.350655 0.6527790  9.697876
## 19        St. Paul    10.619705 0.7666970  9.853008
## 20       San Diego    13.684053 3.0034199 10.680633
## 21         Buffalo    12.563828 0.7078944 11.855933
## 22     San Antonio    21.019638 0.7947887 20.224849
## 23     New Orleans    21.657867 1.0283866 20.629481
```

```
## 24        El Paso    23.714359 1.5316868 22.182672
## 25      Harlingen    30.211050 2.4513559 27.759694
```

```
aggregate_data <- function(data,
                           cols_to_keep,
                           pop_col,
                           agg_noneuca = TRUE){


  pop_cols <- c("pop_ACS", "pop_MPI")

  data_pop <- data %>%
    select(-setdiff(pop_cols, pop_col)) %>%
    rename("pop" = !!sym(pop_col))

  raw_pop <- data_pop %>%
    group_by(state, region) %>%
    summarize(pop = first(pop), .groups = "drop")

  # aggregate regions if reuqested
  if(agg_noneuca){
    data_pop <- data_pop %>%
      mutate(region = ifelse(region == "EUCA", "EUCA", "nonEUCA"))

    raw_pop <- raw_pop %>%
      mutate(region = ifelse(region == "EUCA", "EUCA", "nonEUCA")) %>%
      group_by(state, region) %>%
      summarize(pop = sum(pop), .groups = "drop")
  }

  # checksum value
  checksum <- data_pop %>%
    group_by(state, region, pop) %>%
    slice_head(n=1) %>%
    ungroup() %>%
    group_by(region) %>%
    summarize(pop_checksum = sum(pop), .groups = "drop")

  # find geographic keys
  geo_keys <- intersect(cols_to_keep, c("state", "region"))

  # population map
  pop_map <- data_pop %>%
    group_by(state, region, pop) %>%
    slice_head(n = 1) %>%
    ungroup() %>%
    group_by_at(geo_keys) %>%
    summarize(pop = sum(pop), .groups = "drop")

  # aggregate arrests, join population back in
  data_pop <- data_pop %>%
    group_by_at(cols_to_keep) %>%
    summarize(n_arrests = sum(n_arrests, na.rm = TRUE), .groups = "drop") %>%
    left_join(pop_map, by = geo_keys) %>%
```

```r
    mutate(ratio = n_arrests/pop*1000,
           log_pop = log(pop))

  # checksum
  total_aggregated <- data_pop %>%
    group_by_at(geo_keys) %>%
    summarize(pop = first(pop), .groups = "drop") %>%
    pull(pop) %>%
    sum(na.rm = TRUE)
  total_checksum <- sum(checksum$pop_checksum, na.rm = TRUE)
  if (!(dplyr::near(total_checksum, total_aggregated))) {
    warning("Checksum Failed!")
  }

  return(data_pop)
}
```

MPI, state

```r
trajectory <- read_csv("../data/glm_trajectory_type_threatlevel_convicted_ACS_MPI.csv")
```

```
## Rows: 231336 Columns: 10
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## chr (4): state, region, method, alt_method
## dbl (6): window, convicted, threat_level, n_arrests, pop_ACS, pop_MPI
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
trajectory_agg <- aggregate_data(trajectory, c("region", "state", "window"), "pop_MPI", TRUE) %>%
  mutate(window = ifelse(window < 0, -1, 0)) %>%
  group_by(window, region, state, pop) %>%
  summarize(n_arrests = sum(n_arrests)) %>%
  mutate(ratio = n_arrests/pop *1000)
```

```
## `summarise()` has regrouped the output.
## i Summaries were computed grouped by window, region, state, and pop.
## i Output is grouped by window, region, and state.
## i Use `summarise(.groups = "drop_last")` to silence this message.
## i Use `summarise(.by = c(window, region, state, pop))` for per-operation
##   grouping (`?dplyr::dplyr_by`) instead.
```

```r
trajectory_agg_euca <- trajectory_agg %>%
  filter(region == "EUCA") %>%
  pivot_wider(id_cols = state, names_from = window, values_from = ratio) %>%
  mutate(diff_euca = `0`-`-1`) %>%
  mutate(region = "EUCA") %>%
  select(diff_euca, state)

trajectory_agg %>%
  filter(region == "nonEUCA") %>%
  pivot_wider(id_cols = state, names_from = window, values_from = ratio) %>%
  mutate(diff_noneuca = `0`-`-1`) %>%
  mutate(region = "Non-EUCA") %>%
  select(diff_noneuca, state) %>%
```

```r
  merge(trajectory_agg_euca) %>%
  mutate(diff = diff_noneuca - diff_euca) %>%
  arrange(diff)
```

```
##                    state diff_noneuca diff_euca        diff
## 1               OKLAHOMA    0.3168317  0.000000   0.3168317
## 2            CONNECTICUT    3.7443182  1.363636   2.3806818
## 3                 HAWAII    6.6428571  3.928571   2.7142857
## 4             CALIFORNIA    9.4052990  5.367521   4.0377776
## 5               ILLINOIS    8.6833013  2.909091   5.7742104
## 6           RHODE ISLAND    7.8518519  2.000000   5.8518519
## 7                 ALASKA    8.4444444  1.500000   6.9444444
## 8             WASHINGTON    8.6195286  1.276596   7.3429329
## 9         NORTH CAROLINA   10.9014085  2.541667   8.3597418
## 10                NEVADA   13.5645161  4.500000   9.0645161
## 11              NEW YORK   12.8112450  2.545455  10.2657904
## 12              MARYLAND   14.3631285  3.266667  11.0964618
## 13                OREGON   12.9926471  1.882353  11.1102941
## 14             WISCONSIN   14.2000000  3.000000  11.2000000
## 15            NEW JERSEY   16.8590604  5.620690  11.2383707
## 16         MASSACHUSETTS   14.6676301  2.428571  12.2390586
## 17              COLORADO   17.5188679  2.611111  14.9077568
## 18               FLORIDA   24.5585429  5.859155  18.6993880
## 19                KANSAS   21.7088608  2.750000  18.9588608
## 20               INDIANA   25.1428571  5.500000  19.6428571
## 21              DELAWARE   19.6785714 -1.000000  20.6785714
## 22               GEORGIA   23.9232456  2.750000  21.1732456
## 23              KENTUCKY   25.6964286  3.600000  22.0964286
## 24              VIRGINIA   26.5176471  3.950000  22.5676471
## 25               ARIZONA   26.4068966  2.533333  23.8735632
## 26                 TEXAS   34.6857440  9.177778  25.5079662
## 27                  UTAH   28.8740157  3.300000  25.5740157
## 28               VERMONT   27.0000000  1.000000  26.0000000
## 29          PENNSYLVANIA   31.6018519  5.235294  26.3665577
## 30        SOUTH CAROLINA   28.7322835  1.846154  26.8861296
## 31              NEBRASKA   31.6274510  4.000000  27.6274510
## 32                  IOWA   34.5454545  4.400000  30.1454545
## 33                 IDAHO   35.7272727  5.500000  30.2272727
## 34             LOUISIANA   37.3440860  1.000000  36.3440860
## 35         NEW HAMPSHIRE   40.3750000  3.000000  37.3750000
## 36              MISSOURI   40.7894737  1.875000  38.9144737
## 37              ARKANSAS   43.5967742  3.750000  39.8467742
## 38             TENNESSEE   45.3431953  3.545455  41.7977407
## 39                  OHIO   47.6172840  5.400000  42.2172840
## 40             MINNESOTA   52.7142857  5.500000  47.2142857
## 41              MICHIGAN   51.1587302  1.062500  50.0962302
## 42           MISSISSIPPI   61.7826087  8.000000  53.7826087
## 43  DISTRICT OF COLUMBIA   63.3461538  2.500000  60.8461538
## 44               ALABAMA   71.2121212  8.600000  62.6121212
## 45            NEW MEXICO   70.9333333  2.000000  68.9333333
## 46               MONTANA  131.0000000  2.000000 129.0000000
## 47          NORTH DAKOTA  208.0000000  3.000000 205.0000000
## 48                 MAINE  364.0000000  4.000000 360.0000000
## 49          SOUTH DAKOTA  545.0000000 10.000000 535.0000000
```

```
## 50             WYOMING  840.0000000 22.000000  818.0000000
## 51       WEST VIRGINIA 1973.0000000 25.000000 1948.0000000
```

ACS, state

```
trajectory <- read_csv("../data/glm_trajectory_type_threatlevel_convicted_ACS_MPI.csv")
```

```
## Rows: 231336 Columns: 10
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## chr (4): state, region, method, alt_method
## dbl (6): window, convicted, threat_level, n_arrests, pop_ACS, pop_MPI
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
trajectory_agg <- aggregate_data(trajectory, c("region", "state", "window"), "pop_ACS", TRUE) %>%
  mutate(window = ifelse(window < 0, -1, 0)) %>%
  group_by(window, region, state, pop) %>%
  summarize(n_arrests = sum(n_arrests)) %>%
  mutate(ratio = n_arrests/pop *1000)
```

```
## `summarise()` has regrouped the output.
## i Summaries were computed grouped by window, region, state, and pop.
## i Output is grouped by window, region, and state.
## i Use `summarise(.groups = "drop_last")` to silence this message.
## i Use `summarise(.by = c(window, region, state, pop))` for per-operation
##   grouping (`?dplyr::dplyr_by`) instead.
```

```
trajectory_agg_euca <- trajectory_agg %>%
  filter(region == "EUCA") %>%
  pivot_wider(id_cols = state, names_from = window, values_from = ratio) %>%
  mutate(diff_euca = `0`-`-1`) %>%
  mutate(region = "EUCA") %>%
  select(diff_euca, state)

trajectory_agg %>%
  filter(region == "nonEUCA") %>%
  pivot_wider(id_cols = state, names_from = window, values_from = ratio) %>%
  mutate(diff_noneuca = `0`-`-1`) %>%
  mutate(region = "Non-EUCA") %>%
  select(diff_noneuca, state) %>%
  merge(trajectory_agg_euca) %>%
  mutate(diff = diff_noneuca - diff_euca) %>%
  arrange(diff)
```

```
##                 state diff_noneuca  diff_euca       diff
## 1              HAWAII    0.8680637  1.17370892 -0.3056453
## 2            OKLAHOMA    0.1429044  0.00000000  0.1429044
## 3         CONNECTICUT    1.5634267  0.22448369  1.3389430
## 4              ALASKA    1.7696642  0.42298202  1.3466822
## 5             VERMONT    1.8634826  0.07418948  1.7892932
## 6          CALIFORNIA    2.7408858  0.70138680  2.0394990
## 7            NEW YORK    2.5608119  0.29397329  2.2668386
## 8          WASHINGTON    2.7400603  0.25381033  2.4862500
## 9            ILLINOIS    3.1196902  0.50043527  2.6192549
## 10       RHODE ISLAND    3.2869746  0.23938478  3.0475898
```

```
## 11            WISCONSIN    3.4641046  0.33424322  3.1298614
## 12           NEW JERSEY    4.0341597  0.51959147  3.5145683
## 13               NEVADA    4.7216774  0.73766864  3.9840088
## 14        MASSACHUSETTS    5.1681771  0.40071185  4.7674652
## 15               OREGON    5.4246217  0.37623157  5.0483901
## 16        NEW HAMPSHIRE    5.6295315  0.52230231  5.1072292
## 17             MARYLAND    5.6521084  0.51565378  5.1364547
## 18       NORTH CAROLINA    5.7765505  0.50840536  5.2681451
## 19             MICHIGAN    5.9066735  0.10509397  5.8015795
## 20              FLORIDA    6.8422926  0.74831046  6.0939821
## 21             DELAWARE    5.9228206 -0.17146776  6.0942884
## 22         NORTH DAKOTA    7.4309600  0.43975374  6.9912062
## 23              INDIANA    8.3868490  0.91975167  7.4670974
## 24             COLORADO    8.0669509  0.48859596  7.5783550
## 25                 OHIO    8.1063131  0.46034236  7.6459708
## 26             KENTUCKY    8.4504892  0.61393636  7.8365528
## 27         PENNSYLVANIA    8.8398734  0.94148013  7.8983933
## 28               KANSAS    8.9241112  0.57830819  8.3458030
## 29             VIRGINIA    9.0278174  0.65019506  8.3776224
## 30                 IOWA    9.5709446  0.92514718  8.6457974
## 31              ARIZONA    9.5426672  0.28553180  9.2571354
## 32              GEORGIA   10.3189227  0.57255882  9.7463639
## 33              MONTANA   10.0421617  0.16283993  9.8793218
## 34            MINNESOTA   11.1202808  0.95613928 10.1641415
## 35             MISSOURI   10.8048573  0.27411780 10.5307395
## 36             NEBRASKA   11.5422871  0.68440414 10.8578830
## 37                MAINE   11.9504908  0.17265193 11.7778389
## 38                TEXAS   13.8881945  1.39670472 12.4914897
## 39                IDAHO   13.5812282  1.01033295 12.5708952
## 40                 UTAH   15.3334727  0.71953427 14.6139385
## 41       SOUTH CAROLINA   15.5302369  0.40959126 15.1206457
## 42         SOUTH DAKOTA   18.5475088  1.71408982 16.8334190
## 43            LOUISIANA   17.4556823  0.20524398 17.2504383
## 44             ARKANSAS   19.5532343  1.50142636 18.0518079
## 45            TENNESSEE   21.2641972  0.75773767 20.5064595
## 46          MISSISSIPPI   23.1339031  1.75054705 21.3833561
## 47 DISTRICT OF COLUMBIA   22.9784028  0.55859680 22.4198060
## 48           NEW MEXICO   24.0451977  0.21640338 23.8287944
## 49              ALABAMA   28.5308437  1.57480315 26.9560405
## 50              WYOMING   55.1144938  4.29435877 50.8201350
## 51        WEST VIRGINIA   86.4062363  3.25478453 83.1514518
```